

OSSP:

Commands:

Uname : It displays the System information

Lscpu : It displays the complete CPU information

```
[vigneshkumar@vigneshs-MacBook-Air ~ % ./process
Enter Linux command: sysctl

Child Process
Child PID = 5399
usage: sysctl [-bdeFhlNnoqtWx] [ -B <bufsize> ] [-f filename] name[=value] ...
       sysctl [-bdeFhlNnoqtWx] [ -B <bufsize> ] -a

Parent Process
Parent PID = 5396
Child PID = 5399
vigneshkumar@vigneshs-MacBook-Air ~ %
```

Ps : It displays the running process of our file

```
[vigneshkumar@vigneshs-MacBook-Air ~ % ./process
Enter Linux command: ps

Child Process
Child PID = 5299
  PID TTY          TIME CMD
 5269 ttys000    0:00.03 -zsh
 5298 ttys000    0:00.01 ./process

Parent Process
Parent PID = 5298
Child PID = 5299
vigneshkumar@vigneshs-MacBook-Air ~ %
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among multiple programs by giving each one a turn to execute.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

The operating system gives memory to each program that is running. It makes sure one program does not use another program's memory. Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system stores and organizes files on the storage device. Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.